Navigating the Indian Power Sector in the Face of Extreme Heat

CDRI

ISUW-RESILIENCY OF THE POWER SYSTEMS AGAINST EXTREME WEATHER EVENTS

12/03/2026

Summer and News

India saw 536 heatwave days this summer, warmest June for NW region since 1901: IMD

Urban Heat Islands make nights warmer in the Capital

Delhi experienced its highest-ever peak power demand of 8,656 MW on June 18, 2024, as residents heavily relied on air conditioning during a severe heatwave.

West Bengal: 36% Spike in Power Demand Due to Heat

Context

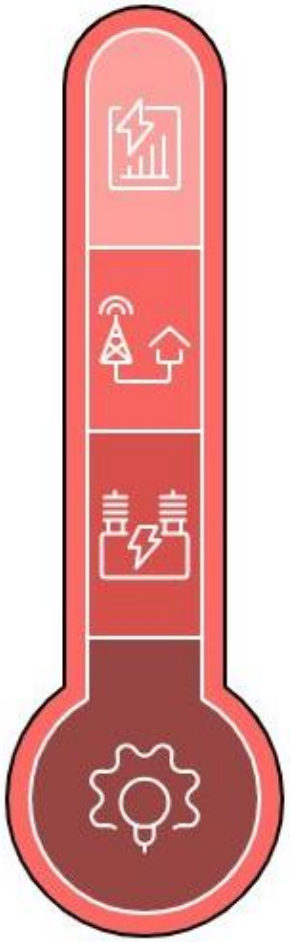
- India is experiencing more intense, frequent, and longer heatwaves
- Urban areas face worse impacts due to the Urban Heat Island effect
- Concurrent hot day & night events are increasing
- The power sector is particularly vulnerable to these climate stressors

Prayas (Energy Group), with CDRI, developed a scoping note identifying extreme heat impacts across the power sector value chain and proposed action ideas with current status and improvement areas.

https://energy.prayaspune.org/our-work/research-report/navigating-the-indian-power-sector-in-the-face-of-extreme-heat-a-scoping-note

https://cdri.world/upload/pages/1830739518738297_202504291220scoping_report.pdf

Key Impacts on Power Sector



Power Sector Element	Impact of Extreme Heat
Electricity Demand	- Sharp surges in peak demand due to increased cooling load - Stress on short-term planning
Transmission & Distribution	- Overheating of transformers, cables, substations - Risk of equipment failure and outages
Generation	- Reduced thermal plant efficiency - Lower hydro output due to water stress-Variability in solar/wind
System Operation	- Greater difficulty balancing supply and demand - Higher probability of cascading failures

Planning & Response Priority Areas

Update Standards (BIS/IS)
Review, Revise & Consolidate construction, operation, and equipment standards to ensure heat resilience.

Grid Resilience
Improve real-time balancing, system reserves, and incorporate storage solutions.

Demand-Side Management/ Flexibility
Scale demand response programs, efficient cooling appliances, and energy efficiency codes.

Cost Benefit Analysis for Resilience
Build cost-benefit frameworks to justify investments for heat-adaptive infrastructure.

Strengthening Resilience

Immediate Actions

- Consolidate 2024/2025 responses and prepare next summer action plan (CEA/Discoms)
- Organize regional heat audit workshops (CDRI, NDMA, Regulators)
- Develop heat emergency protocols: targeted load shedding/shifting (Discoms)
- Public awareness campaigns for demand moderation (like Kerala, May 2024/2025)

Medium-Term Actions

- Improve granular demand forecasting, especially for heat-driven surges
- Enforce ECBC & enhance cooling appliance efficiency
- Plan and implement Demand Response & seasonal ToD tariffs
- Asset health monitoring, remote metering & heat-specific O&M

Realities of the Power Sector

- Rs 8.75 lakh cr annual expenses (incl. subsidy), FY 23.
- Rs 6.47 lakh crore accumulated losses as per balance sheet, FY 23.
- Increasingly subsidized (agriculture and now residential)
  - ~ Rs 1.69 lakh crore (FY 23), around 17.5% of total expenses; Hence highly politicized sector
- Regulated sector
  - Generation – delicensed, Transmission and Distribution needs licenses
  - Generation and Transmission is increasingly having larger private participation, Distribution is largely state owned except for some city pockets (Delhi, Mumbai, Ahmedabad, Kolkata)
  - Weakening governance architecture over time, in spite of increasing complexity in the sector.
- Short term financial crisis led operations, lower focus on long term optimal economic/fiscal prudence. Lack of trust between consumers & DISCOMs, esp. wrt to metering, billing and lack of supply reliability.
- Increasing electrification of the economy, Push to increase renewables, Relatively small, but growing power markets. Lack of emphasis on demand side resources/energy efficiency.

Way Forward

Institutionalize climate resilience across power planning

- Shift from reactive to proactive planning
- operationalize climate resilience

Foster multi-sector collaboration (climate, energy, urban development)

- Align power planning with weather, urban, and climate policies

Prioritize public data sharing, M&V of resiliency measures

- Share resiliency plans, data, and outcomes publicly

Ensure a reliable, efficient, and heat-resilient power future

- Improve reliability and equity in electricity access during stress events

Thank you